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Editors
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Dear Editors,

Please consider for publication my manuscript, "Exact Diffusion Design for Maximally Collective Stable Turing Patterns in Binary-Complex Mass-Action Networks: Topology-Wide All-Spectrum Localization and Exponent-Optimal Stationary Heterogeneity Trade-Offs."

The paper addresses a structural question in multicomponent reaction-diffusion systems: how small can a principal subsystem responsible for instability be? For every $n \geq 4$, it constructs one sparse classical mass-action topology for which every principal subsystem below order $n-1$ is Hurwitz throughout every positive realization, while an order-$(n-1)$ subsystem is unstable. The same topology admits an exact primary supercritical Turing bifurcation with positive patterns that are locally exponentially asymptotically stable in the fixed-integrated-mass $H^1$ phase space. The paper also derives a necessary-and-sufficient stationary diffusion law, sharp topology-specific heterogeneity lower bounds, and an exact stable family in which diffusion and equilibrium concentration contrasts are both of square-root order, with the exponent optimal among stationary-crossing realizations of this topology.

The manuscript is self-contained and distinguishes its principal-block theorem from projected-network injectivity, parameter-rich kinetic cores, and fixed-Jacobian criteria. The construction is synthetic, and the nonlinear conclusions are local for each fixed dimension. Exact symbolic certificates, independent verification code, and numerical illustration data accompany the submission. The target immutable release is version 1.0.10 at GitHub; archived versions share the concept DOI 10.5281/zenodo.21753404, and the exact preceding version 1.0.9 is independently addressable at 10.5281/zenodo.22478273.

The manuscript is not under consideration by, and has not been submitted to, another journal. The author received no specific funding for this work and declares no competing interests.

Sincerely,

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